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Modern Cipher and Module Objectives

Modern Cipher and Block Cipher (vs. Stream Cipher)

Ideal Block Cipher

Fistel Cipher

Data Encryption Standard (DES)

Video: DES Overview3 min

Video: DES Round Function1 min

Reading: About Pseudocode30 min

Peer-graded Assignment: DES Overview Pseudo-Code15 min

Review Your Peers' DES Overview Pseudo-Code

Video: DES Subkey Generation1 min

Video: DES Security4 min

Reading: Lecture Slides for Block Cipher and DES15 min

Quiz: DESSubmitted

Peer-graded Assignment: DES Overview Pseudo-Code

You passed!
Congratulations! You earned 25 / 25 points. Review the feedback below and continue the course when you are ready.

InstructionsMy submissionDiscussions

Some of your fellow learners have submitted their reviews anonymously. All names are still visible to course instructors.

Pseudocode for DES Encryption

Submitted on July 12, 2024

Shareable Link

OVERALL ASSIGNMENT RUBRIC

Does the pseudo-code correctly describe DES cipher operation?

0 points

Most, if not all, of the pseudo-code components are flawed.

2 points

The answer has some components but requires major overhaul, or the answer built another cipher.

4 points

The answer misses a few steps/variables, impacting its logic, or only works in special circumstances (e.g., restrictions on variables).

6 points

The answer misses a step/variable but the error is relatively minor and can be corrected quickly.

8 points

The answer provides a correct DES encryption supporting the correct data and key.

F

How is the pseudo-code presentation?

0 points

I cannot follow the pseudo-code logic.

1 point

The pseudo-code is very difficult to follow, for example, it is so long that it is distracting, or the documentation is poor.

2 points

The pseudo-code is hard to follow but is understandable. Or the pseudo-code is unnecessarily long.

3 points

The pseudo-code is generally easy to follow but lacks some details that can help its understanding and readability.

4 points

The pseudo-code is easy to follow with clear definitions and documentation.

F

PROMPT

Which programming language, if any, are you familiar with?

I am familiar with python, R, java, C, C++. However I shall use python to implement the pseudocode.

RUBRIC

Are you (the reviewer) familiar with the programming language that the submitter identified in his/her answer? (The submitter receives credit regardless.)

1 point

No

1 point

Yes

F

PROMPT

Write your answer for the pseudo-code for the DES encryption.

Pseudo-code for DES Encryption:

Constants / Design Parameters for DES as a Feistel Cipher

- BLOCK_SZ = 64 # length of block
- KEY_SZ = 56 # length of key
- NUM_ROUNDS = 16 # number of rounds

- subkey_gen_fn: generate_subkeys # generates 16 different subkeys of size 48 bits from input key
- round_fn: F # the round function

Input:

- plaintext: input text of length BLOCK_SZ (64-bits) to be encrypted
- key: input key of length KEY_SZ (56-bits) to be used for subkey generation

Output:
- ciphertext: encrypted output of length BLOCK_SZ (64-bits)

Variables:

- subkeys[16]: array of 48-bit subkeys for each of the 16 rounds
- left[0..15], right[0..15]: arrays to hold the left and right halves of the data during the rounds
- temp: temporary variable used for swapping

Functions (helpers)

- IP (the initial permutation, taking 64 bits of input and outputting 64 bits)
- F (the round function, taking 32 bits of data and 48 bits of subkey and outputting 32 bits)
- FP (the final permutation, taking 64 bits of input and outputting 64 bits)
- generate_subkeys (function to generate 16 subkeys each of length 48 bits from input key of length 56 bits)

Steps:

0. Generate 16 subkeys each of length 48 bits from the input key with a subkey generation algorithm

subkeys = generate_subkeys(key) # the function returns array of 16 different 48-bit subkeys

1. Apply the initial permutation (IP) to the plaintext

data = IP(plaintext)

2. Split the permuted data into two equal halves

left[0] = data[0..31] # First 32 bits
right[0] = data[32..63] # Last 32 bits

3. Perform 16 rounds of processing

for round from 1 to NUM_ROUNDS do: # start for loop

 # a. Apply the round function (F) to the right half and the round's subkey

 temp = F(right[round-1], subkeys[round-1])

 # b. XOR the output of the round function with the left half

 temp = temp XOR left[round-1]

 # c. The left half for the next round is the right half of the current round

 left[round] = right[round-1]

 # d. The right half for the next round is the result stored in temp

 right[round] = temp

 # end for loop

4. After the final round, combine the halves in reverse order (right[16], left[16])

combined_data = right[16] + left[16]

5. Apply the final permutation (FP) to the combined data

ciphertext = FP(combined_data)

6. Return the ciphertext as the output

Python Implementation

A python function implementing the above pseudocode will look like the following one:

def DES_encrypt(plaintext, key):
 ...

 # Encrypts the given plaintext using the DES algorithm.

 Parameters:
 plaintext (str): The plaintext to be encrypted.
 key (list): The key to be used for subkey generation for DES.

 Returns:
 str: The ciphertext generated from the encryption process.
 ...

 # Generate subkeys from the input key
 subkeys = generate_subkey(key)

 # Apply #
 permuted_data = IP(plaintext) # initial permutation
 left, right = [0] * (NUM_ROUNDS+1), [0] * (NUM_ROUNDS+1)
 left[0], right[0] = permuted_data[32], permuted_data[64]

 # 16 rounds
 for round in range(1, NUM_ROUNDS+1):
 temp = XOR(right[round-1], subkeys[round-1])
 left[round-1], right[round-1] = temp, temp

 # Combine and apply FP
 combined_data = right[NUM_ROUNDS] + left[NUM_ROUNDS] # NUM_ROUNDS = 16
 ciphertext = FP(combined_data) # final permutation to obtain the output ciphertext
 return ciphertext

Does the pseudo-code correctly describe DES Encryption supporting correct data/key types/lengths?

0 points

Most, if not all, of the pseudo-code components are flawed.

2 points

The answer has some components but requires major overhaul or built something else (not a DES cipher).

4 points

The answer misses a few steps/variables, impacting its logic, or only works in special circumstances (e.g., restrictions on variables).

6 points

The answer misses a step/variable but it is minor, and the pseudo-code still implements DES Encryption or the error can be corrected quickly.

8 points

The answer provides a correct DES encryption supporting the correct input/output and arbitrary length of key.

F

How is the pseudo-code presentation?

0 points

I cannot follow the logic of the pseudo-code.

1 point

The pseudo-code is so long that it is distracting; and the documentation is poor.

2 points

The pseudo-code is hard to follow but is understandable. Or the pseudo-code is unnecessarily long.

3 points

The pseudo-code is generally easy to follow but lacks some documentation.

4 points

The pseudo-code is easy to follow with clear definitions and explanations/documentation.

F

Start new attempt

Comments

Comments left for the learner are visible only to that learner and the person who left the comment.

SD

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